EXPLANATIONS

1. (*) The electrostatic force

$F_e = k\frac{|q_A q_B|}{r^2}$

the gravitational force

$F_g = G\frac{m_A m_B}{r^2}$

the ratio of the magnitudes of the electrostatic force to the gravitational force:

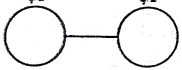
$\frac{F_e}{F_g} = \frac{k|q_A q_B|}{G m_A m_B}$

$\frac{9\times10^9\times6.67\times10^{-19}\times9.6\times10^{-10}}{6.67\times10^{-11}\times19.2\times10^{-27}\times9\times10^{-27}}$

$\frac{1}{2}\times10^{45}$

Charge is not an integral multiple of electron.

2. (*d*) Charge q will be divided equally on each sphere.



$9\times10^{-3} = \frac{9\times10^9\times(4\times10^{-8}\times4\times10^{-8})}{4\times r^2}$

Solving, we get

$\Rightarrow r = 2\times10^{-2}\,m \quad \Rightarrow 2\,cm$

3. (*b*) In air, force is $F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r_2}$

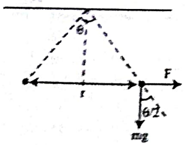
In medium

$F' = \frac{1}{4\pi(K\varepsilon_0)}\frac{q_1q_2}{\left(\frac{r}{\ }\right)^2} = \frac{25}{4\pi(5\varepsilon_0)}\frac{q_1q_2}{(r)^2} = 5F$

(Take density of water = 1 g/cc)

4. [3] In air,

$\tan\frac{\theta}{2} = \frac{F}{mg} = \frac{q^2}{4\pi\varepsilon_0 r^2 mg}$ ...(*i*)



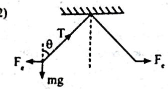
In water,

$\tan\frac{\theta}{2} = \frac{F'}{mg'} = \frac{q^2}{4\pi\varepsilon_0\varepsilon_r r^2 mg_{eff}}$ ...(*ii*)

Equating, both equations

$\Rightarrow \varepsilon_0\, g = \varepsilon_0\varepsilon_r g\left[1-\frac{1}{1.5}\right] \Rightarrow \varepsilon_r = 3$

5. (2)



$T\cos\theta = mg$

$T\sin\theta = F_e$

$\tan\theta = \frac{F_e}{mg}$

$\tan\theta = \frac{F_e}{\rho_B Vg}$ ...(*i*)

$\tan\theta = \frac{\frac{F_e}{k}}{(\rho_B-\rho_L)Vg}$ ...(*ii*)

From Eq. (*i*) & (*ii*)

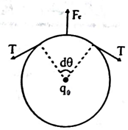
$\rho_B Vg = (\rho_B-\rho_L)kVg$

$1.4 = 0.7\,k$

the dielectric constant of the liquid is

$k = 2$

6. [3]



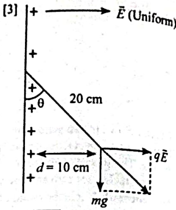
$\Rightarrow 2T\sin\frac{d\theta}{2} = \frac{kq_0}{R^2}\cdot\lambda R d\theta$

$\Rightarrow T = \frac{Kq_0Q}{(R^2)\times2\pi}$

$= \frac{(9\times10^9)(2\pi\times30\times10^{-12})}{(0.30)^2\times2\pi}$

$= \frac{9\times10^{-1}\times30}{9\times10^{-2}} = 3N$

7. [3]



$\boxed{\tan\theta = \frac{qE}{mg}}$

$\tan30^\circ = \frac{q\times2\times10^4}{1\times10^{-3}\times10}$

$\boxed{\frac{1}{\sqrt{3}} = q\times10^6}$

$\boxed{q = \frac{1}{\sqrt{3}}\times10^{-6}C}$

$x = 3$

8. (*a*) Coulombian force

$F_e = \frac{kQ_1Q_2}{r^2}$

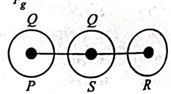
$= \frac{9\times10^9\times1.6\times10^{-19}\times1.6\times10^{-19}}{r^2}$

$F_e = \frac{Gm_1m_2}{r^2}$

$= \frac{6.67\times10^{-11}\times9.1\times10^{-31}\times1.6\times10^{-27}}{r^2}$

$\frac{F_e}{F_g} \cong 0.23\times10^{40} \cong 2.3\times10^{39}$

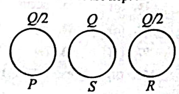
9. (2)



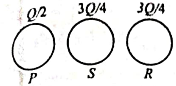
$F_{ps} \propto Q^2$

$F_{PS} = 16N$

When P and R are kept in contact then



When S and R are brought in contact then



New force between P and S is:

$F_{PS} \propto \frac{Q}{2}\times\frac{3Q}{4}$

$F_{PS} \propto \frac{3Q^2}{8} = \frac{3}{8}\times16 = 6N$

10. (*a*) Force between the charges in medium,

$F = \frac{1}{(4\pi\varepsilon_0)}\frac{q_1q_2}{kd^2}$

Force between the charges in air,

$F_{Air} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{d'^2}$

Given, $F = F_{Air}$

$\Rightarrow \frac{q_1q_2}{4\pi\varepsilon_0 kd^2} = \frac{q_1q_2}{4\pi\varepsilon_0 d'^2} \quad d' = d\sqrt{k}$

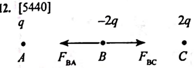
11. (*b*) Let the charge on first and second part be x & $10-x$ respectively.

$F = \frac{Kx(10-x)}{r^2}$

For F to be maximum, $\frac{dF}{dx} = 0$

$= \frac{K}{r^2}(10-2x) = 0 \quad x = \frac{10}{2} = 5\mu C$

12. [5440]



Force on B due to A

$F_{BA} = \frac{Kq(2q)}{\left(\frac{3}{4}R\right)^2} = \frac{K32q^2}{9R^2}$

Force on B due to C

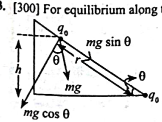
$F_{BC} = \frac{K(2q)(2q)}{\left(\frac{R}{4}\right)^2} = \frac{64Kq^2}{R^2}$

Net force on B

$F_B = F_{BC} - F_{BA} = \frac{544Kq^2}{9R^2}$

$= \frac{544\times9\times10^9\times(2\times10^{-6})^2}{9\times(2\times10^{-2})^2} = 5440\,N$

13. [300] For equilibrium along the plane



$mg\sin\theta = K\times\frac{q_0^2}{(h\,\mathrm{cosec}\,\theta)^2}$

Put θ 30°

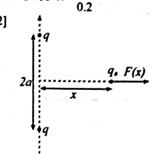
$\therefore h^2 = \frac{1}{K}\times\frac{q_0^2}{mg\,\mathrm{cosec}30^\circ}$

$= 9\times10^9\times\frac{(2\times10^{-6})^2}{0.02\times10\times2}$

$\therefore K = \frac{1}{4\pi E_0} = 9\times10^9\,N\,m^2/C^2$

$\therefore h = 3\times10^4\times\frac{2\times10^{-6}}{0.2} = 0.3\,m = 300\,mm$

14. [2]



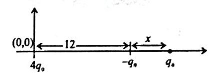
Force acting on the charge q_0,

$F = \frac{2Kqq_0x}{(x^2+a^2)^{3/2}}$

For F to be maximum

$\frac{dF}{dx} = 0 \Rightarrow x = \frac{a}{\sqrt{2}}$

15. [24]



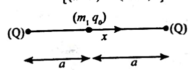
As the net force on the proton is zero

$\Rightarrow \frac{q_0}{x^2} = \frac{4q_0}{(x+12)^2} \Rightarrow x = 12$

$\Rightarrow$ Distance of the proton from origin $= x + 12 = 24\,cm.$

16. (*a*) Resultant force on charge q_0

$F_R = \left\{\frac{kQq_0}{(a-x)^2} - \frac{kQq_0}{(a+x)^2}\right\}$



$= -kQq_0\left\{\frac{a^2+x^2+2ax-a^2-x^2+2ax}{(a^2-x^2)^2}\right\}$

$F_R = -\frac{kQq_0(4ax)}{(a^2-x^2)^2}$

From Newtons second law, $a = \frac{F}{m}$

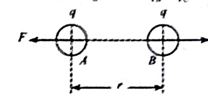
$\therefore a = -\frac{kQq_0\cdot4a}{a^4m}x = -\left(\frac{4kQq_0}{a^3m}\right)x$

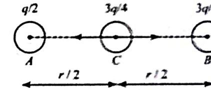
$T = 2\pi\sqrt{\frac{a^3m}{4kQq_0}} = \sqrt{\frac{4\pi^2a^3m\times4\pi\varepsilon_0}{4Qq_0}}$

$= \sqrt{\frac{4\pi^3\varepsilon_0 ma^3}{Qq_0}}$

17. (*b*) For sphere '*C*' after contacting with '*A*', $q_A = q_C = \frac{q}{2}$.

After contacting with '*B*', $q_B = q_C = \frac{3q}{4}$





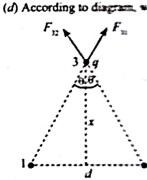
Initially, $F = \frac{kq^2}{r^2}$

$F_1 = \frac{9kq^2}{4r^2} = \frac{9}{4}F$

$F_2 = \frac{3}{2}\frac{kq^2}{r^2} = \frac{3}{2}F$

$\therefore F_{Net} = \frac{9}{4}F - \frac{3}{2}F = \frac{3}{4}F$

18. (*d*) According to diagram, we can write



$F_{31} = F_{32} = \frac{KqQ}{x^2+\left(\frac{d}{2}\right)^2}$

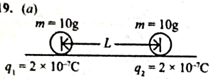
$\Rightarrow F = 2F_{31}\cos\theta = \frac{2KqQx}{\left[x^2+\left(\frac{d}{2}\right)^2\right]^{\frac{3}{2}}}$

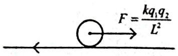
F is net force on q

$\because$ For maxima of force, $\frac{dF}{dx} = 0$

On solving, we get $x = \frac{d}{2\sqrt{2}}$

19. (a)





$f_s = \mu N = \mu mg$

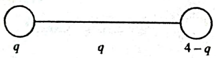
According to the question, Coulumb force an each charge = Frictional force

$\Rightarrow \mu mg = \frac{kq_1q_2}{L^2}$

$\Rightarrow 0.25\times0.01\times10 = \frac{9\times10^9\times2\times10^{-7}\times2\times10^{-7}}{L^2}$

$\Rightarrow L = 12$ cm

20. (b)

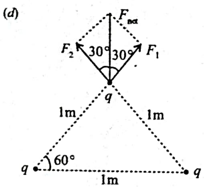


$F = \frac{kq(4-q)}{d^2}$

$\frac{dF}{dq} = 0 \Rightarrow \frac{k}{d^2}(4-2q) = 0$

$q = 2$

21. (d)

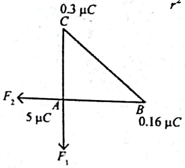


$F_1 = F_2 = \frac{kq^2}{1^2}$

$F_{net} = 2F\cos30°$

$\therefore \frac{F_{net}}{F_1} = 2\cos30° = 2\times\frac{\sqrt3}{2} = \sqrt3:1$

22. [17] Electrostatic force, $F = \frac{KQ_1Q_2}{r^2}$



$F_1 = \frac{9\times10^9\times5\times10^{-6}\times0.3\times10^{-6}}{9\times10^{-4}} = 15N$

$F_2 = \frac{9\times10^9\times5\times10^{-6}\times0.16\times10^{-6}}{9\times10^{-4}} = 8N$

$\therefore F_{net} = \sqrt{(15)^2+(8)^2} = 17N$

23. (b) Distance between the charges $r = \ell\sin\theta + \ell\sin\theta$

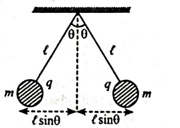
Applying Newton's second law,

$T\sin\theta = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(2\ell\sin\theta)^2}$ ...(i)

$T\cos\theta = mg$ ...(ii)

$\therefore \tan\theta = \frac{q^2}{4\pi\varepsilon_0 mg4\ell^2\theta^2}$

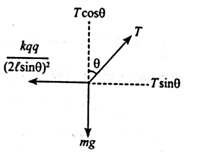
[For small angle, $\tan\theta \approx \theta$]



So, $\theta^3 = \frac{q^2}{16\pi\varepsilon_0 mg\ell^2}$

$\theta = \left(\frac{q^2}{16\pi\varepsilon_0 mg\ell^2}\right)^{1/3}$

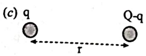
Also separation = $2\ell\sin\theta \cong 2\ell\theta$



$= 2\ell\left(\frac{q^2}{16\pi\varepsilon_0 mg\ell^2}\right)^{1/3} = \left(\frac{8q^2\ell^3}{16\pi\varepsilon_0 mg\ell^2}\right)^{1/3}$

$= \left(\frac{q^2\ell}{2\pi\varepsilon_0 mg}\right)^{1/3}$

24. (c)



The electrostatic repulsion force between q and Q − q.

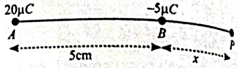
$F = \frac{q(Q-q)}{4\pi\varepsilon_0 r^2} \Rightarrow \frac{dF}{dq} = \frac{1}{4\pi\varepsilon_0 r^2}(Q-2q)$

For maxima of force, $\frac{dF}{dq} = 0$

On differentiating F w.r.t q

$\frac{dF}{dq} = \frac{1}{4\pi\varepsilon_0 r^2}(Q-2q) = 0 \Rightarrow q = \frac{Q}{2}$

25. (d) From observation, we can say that net force on the charge particle at point P is zero.

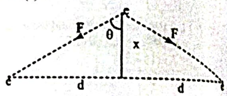


So, $\frac{k(5\mu C)q}{x^2} = \frac{k(20\mu C)q}{(5+x)^2}$

$\Rightarrow \frac{1}{x^2} = \frac{4}{(5+x)^2} \Rightarrow \left(\frac{5+x}{x}\right)^2 = 2^2$

$\Rightarrow \frac{5+x}{x} = 2 \Rightarrow x = 5$ cm

26. (c)

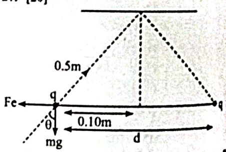


os $\theta = 2\times\frac{ke^2\times x}{(x^2+d^2)^{3/2}}$

$\therefore x << d$

$\therefore F_{net} = \frac{2ke^2}{d^3}x.$

27. [20]



From FBD, $T\sin\theta = kq_1/d_2$ and $T\cos\theta$ = mg Dividing them,

Mg $\tan\theta = kq^2/d^2$

$q = \sqrt{(mg\tan\theta)/25k}$

$= \sqrt{(10\times10^{-6}\times10\times1/\sqrt{24(25)9\times10^9})}$

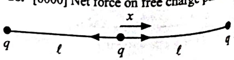
$= \sqrt{(10^{-4}\times4/\sqrt{24\times25\times4\times9\times10^9})}$

$= \sqrt{(10^{-4}/\sqrt{24\times10^{11}})}$

Thus $(a/21)\times10^{-8} = \frac{2}{3}\sqrt{(10^{-15}/\sqrt{24})}$

$a = 20$

28. [6000] Net force on free charge particle.



$F_{net} = \frac{kq^2}{(\ell+x)^2} - \frac{kq^2}{(\ell-x)^2}$

$= kq^2\left[\frac{(\ell-x)^2-(\ell+x)^2}{(\ell^2-x^2)^2}\right] = \frac{kq^2}{\ell^4}(-4\ell x)$

$[x << \lambda]$

From Newton's II-law, $F_{net} = ma$

$\therefore ma = \frac{kq^2}{\ell^4}(-4\ell x) = -\frac{4kq^2}{\ell^3}x$

$a = -\frac{4kq^2}{m\ell^3}x$ Comparing with $a = -\omega^2x$.

$\omega = \sqrt{\frac{4kq^2}{m\ell^3}} = \sqrt{\frac{4\times9\times10^9\times10}{1\times10^{-6}\times1}}$

6×10^8 rad/s = 6000×10^5 rad/s

29. [36] $q = \frac{(2.1-0.1)}{2}nC = 1nC$

(final charge on both sphere)

$F = \frac{9\times10^9\times10^{-18}}{(0.5)^2} = 36\times10^{-9}\left(F = k\frac{q_1\times q_2}{r^2}\right)$

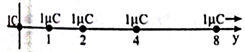
30. (a) $neE = \frac{4}{3}\pi r^3\cdot\rho g$ [$\because q = ne$]

$\Rightarrow n = \frac{4\pi r^3\rho g}{3eE}$

$= \frac{4\times3.14\times(2\times10^{-3})^3\times3\times10^3\times9.81}{3\times1.6\times10^{-19}\times3.55\times10^5}$

$= 1.73\times10^{10}$

31. [12]



$F = k(1C)(1\mu C)\left[1+\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{8^2}+...\right]$

$= 9\times10^3\left[\frac{1}{1-\frac{1}{4}}\right] = 12\times10^3 N$

32. (c) For spherical shell

$E = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$ $(r > R)$

$= 0$ $(r < R)$

Force on charge in electric field

$F = qE$

33. (c) $E\cdot4\pi r^2 = \frac{\int_0^r (kr)\cdot4\pi r^2dr}{\varepsilon_0}$

$\Rightarrow E = \frac{kr^2}{4\varepsilon_0}$ ...(i)

Now, $2Q = \int_0^R kr\cdot4\pi r^2dr$

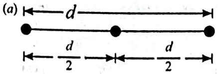
$\Rightarrow K = \frac{2Q}{\pi R^4}$ ...(ii)

$E = \frac{Qr^2}{2\pi\varepsilon_0R^4}$ [From (i) and (ii)]

$\because F_{net} = 0 \Rightarrow \frac{Q2}{4\pi\varepsilon_0(2a)^2} = EQ$

$a = R\cdot8^{-1/4}$.

34. (a)



$\vec F_1 + \vec F_2 = 0$

$\Rightarrow \vec F_1 = -\vec F_2$ ...(i)

$\frac{KQq}{\left(\frac{d}{2}\right)^2} = \frac{-KQQ}{d^2} \Rightarrow q = \frac{-Q}{4}$

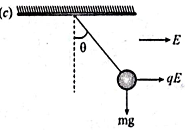
35. (b)

$Q = \int\rho4\pi r^2dr = \int_0^R\left(\frac{A}{r^2}e^{-\frac{2r}{a}}\right)(4\pi r^2)dr$

$= 4\pi A\frac{a}{2}\left(1-e^{\frac{-2R}{a}}\right)$

$R = \frac{a}{2}\log\left(\frac{1}{1-\frac{Q}{2\pi aA}}\right)$

36. (c)



$\tan\theta = \frac{qE}{mg} = \frac{5\times10^{-6}\times2000}{2\times10^{-3}\times10}$

$\tan\theta = \frac{1}{2} \Rightarrow \theta = \tan^{-1}(0.5)$

37. (b,d) General equation of SHM

$x = A\sin(\omega't\pm\phi)$

We know, $\omega = \frac{2\pi}{T}$

Given, $T = \frac{\pi}{\omega}$

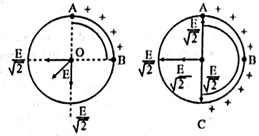
$\therefore \omega' = \frac{2\pi}{\frac{\pi}{\omega}} = 2\omega$

∴ Equation becomes,

$x = a\sin(2\omega t\pm\phi)$

Here coefficient of t is 2ω.

38. (b) The electric field at 'O' due to arc AB is E. Due to symmetry, the electric field components from the additional arc BC cancel out along the diameter AC.



Thus, the magnitude of the electric field at 'O' due to arc ABC is

$E_{net} = \frac{E}{\sqrt2}+\frac{E}{\sqrt2} = \frac{2E}{\sqrt2} = \sqrt2E$

39. (c)

$|E_A| > |E_B|$ since $r_A < r_B$

$\Rightarrow |E_A| = \left|+\frac{\sigma}{2\varepsilon_0}-\frac{\sigma}{2\varepsilon_0}-\frac{1}{4\pi\varepsilon_0}\frac{Q}{r_A^2}\right|$

$= \frac{Q}{4\pi\varepsilon_0r_A^2}$

$\Rightarrow |E_B| = \left|+\frac{\sigma}{2\varepsilon_0}-\frac{\sigma}{2\varepsilon_0}-\frac{1}{4\pi\varepsilon_0}\frac{Q}{r_B^2}\right|$

$= \frac{Q}{4\pi\varepsilon_0r_B^2}$

$\Rightarrow |E_C| = \left|-\frac{\sigma}{2\varepsilon_0}-\frac{\sigma}{2\varepsilon_0}-\frac{1}{4\pi\varepsilon_0}\frac{Q}{r_C^2}\right|$

$= \frac{\sigma}{\varepsilon_0}+\frac{Q}{4\pi\varepsilon_0r_C^2}$

$\Rightarrow |E_D| = \left|\frac{\sigma}{2\varepsilon_0}+\frac{\sigma}{2\varepsilon_0}+\frac{Q}{4\pi\varepsilon_0r_D^2}\right|$

$= \frac{\sigma}{\varepsilon_0}+\frac{Q}{4\pi\varepsilon_0r_D^2}$

$|E_C| \neq |E_D|$ as $r_C \neq r_D$

40. (a) Charge density (σ) is inversely proportional to the radius of curvature (ROC) of the surface: $\sigma \propto \frac{1}{ROC}$

Points with smaller ROC (sharper edges) have higher charge density, while points with larger ROC (flatter regions) have lower charge density.

The order of radius of curvature (ROC) is: $(ROC)_1 < (ROC)_3 < (ROC)_2 = (ROC)_4$

Thus, the charge density (σ) follows:

$\sigma_1 > \sigma_3 > \sigma_2 = \sigma_4$

(A): $\sigma_1 > \sigma_3$ and $\sigma_2 = \sigma_4$ → True (matches the derived order).

(B): $\sigma_1 > \sigma_2$ and $\sigma_3 > \sigma_4$ → True (since $\sigma_1 > \sigma_2 = \sigma_4$).

(C): $\sigma_1 > \sigma_3 > \sigma_2 = \sigma_4$ True (exact match with the solution).

(D): Incorrect order ($\sigma_1 < \sigma_3$ is false).

(E): All densities equal → False (varies with ROC).

41. (*a*) The outer metal body of an aircraft shields individuals inside from lightning strikes by conducting the charge externally. This is due to the electric field being zero inside a conductor's cavity, as per Gauss's law, where charges distribute on the surface. This shielding effect prevents internal damage or harm.

42. (*c*) The electric field $E = \dfrac{KQz}{(z^2+R^2)^{3/2}}$

where $R = a\sqrt{2}$. For maximum

$E, \dfrac{dE}{dz} = 0$ yields $z = \dfrac{R}{\sqrt{2}}$.

Substituting $R = a\sqrt{2}, z = \dfrac{a\sqrt{2}}{\sqrt{2}} = a$.

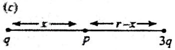
43. (*b*) A charge $+q$ is at the origin (0,0,0), and a charge $+9q$ is at (d,0,0). The point where the electric field is zero lies between them. Equating field magnitudes,

$$\frac{kq}{x^2} = \frac{k \cdot 9q}{(d-x)^2},$$

$$\Rightarrow \frac{1}{x^2} = \frac{9}{(d-x)^2}.$$

$$\Rightarrow \frac{d-x}{x} = 3 \Rightarrow x = \frac{d}{4}.$$

44. (*c*)



$(\vec{E}_{net})_P = 0$

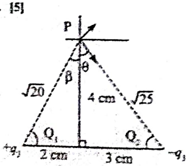
$$\frac{kq}{x^2} = \frac{k \cdot 3q}{(r-x)^2}$$

$$(r-x)^2 = 3x^2$$

$$r - x = \sqrt{3}x$$

$$x = \frac{r}{\sqrt{3}+1}$$

45. [5]



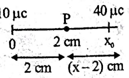
since net electric field at point P is zero so,

$$\frac{Kq_2}{20}\cos\beta = \frac{Kq_3}{25}\cos\theta$$

$$\frac{Kq_2}{20}\frac{4}{\sqrt{20}} = \frac{Kq_3}{25}\frac{4}{\sqrt{25}}$$

$$\frac{q_2}{q_3} = \frac{20}{25}\sqrt{\frac{20}{25}} = \frac{8}{5\sqrt{x}} \Rightarrow 7 \Rightarrow x = 5$$

46. (*a*) 10 μc



For the electric field to be zero at $x = 2$ cm from the 10μc charge.

$$E_P = \frac{K \times 10}{2^2} - \frac{K \times 40}{(x-2)^2} = 0$$

$$\frac{1}{2} = \frac{2}{x-2} \Rightarrow x = 6 \text{ cm}$$

47. (*b*) Electric field inside material of conductor is zero.

48. [6] $E_P = \dfrac{\lambda}{2\pi\varepsilon_0 r_P} - \dfrac{\sigma}{2\varepsilon_0}$

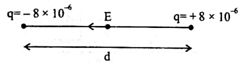
$$= \frac{1}{2\varepsilon_0}\frac{\lambda}{3} - \sigma \qquad \left[\text{For, } r_P = \frac{3}{\pi}\right]$$

$$E_Q = \frac{\lambda}{2\pi\varepsilon_0 r_Q} - \frac{\sigma}{2\varepsilon_0}$$

$$= \frac{1}{2\varepsilon_0}\left|\frac{\lambda}{4} - \sigma\right| \qquad \left[\text{For, } r_Q = \frac{4}{\pi}\right]$$

$$\frac{E_P}{E_Q} = \frac{4}{3}\left(\frac{\lambda - 3\sigma}{\lambda - 4\sigma}\right) = \frac{4}{3}\left(\frac{2\sigma - 3\sigma}{2\sigma - 4\sigma}\right) = \frac{4}{6}$$

49. (*b*)

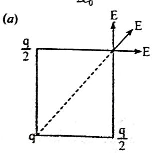


$$E = \frac{k \cdot q}{\left(\frac{d}{2}\right)^2} \times 2 \Rightarrow d^2 = \frac{8\,kq}{E}$$

$$d^2 = \frac{8 \times 9 \times 10^9 \times 8 \times 10^{-6}}{6.4 \times 10^4} \Rightarrow d = 3 \text{ m}$$

50. (*c*) $E_1 = E_2 = \dfrac{\sigma}{2\varepsilon_0}$

51. (*a*)



$$E = \frac{1}{4\pi\varepsilon_0}\frac{(q/2)}{a^2}$$

$$E_{net} = \sqrt{2}E + E$$

$$= \frac{q}{4\pi\varepsilon_0 a^2}\left(\frac{\sqrt{2}+1}{2}\right)$$

52. (*a*) $neE = \dfrac{4}{3}\pi r^3 \cdot \rho g$ [∵ $q = ne$]

$$\Rightarrow n = \frac{4\pi r^3 \rho g}{3eE}$$

$$= \frac{4 \times 3.14 \times (2\times10^{-3})^3 \times 3 \times 10^3 \times 9.81}{3 \times 1.6 \times 10^{-19} \times 3.55 \times 10^5}$$

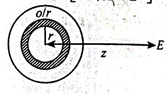
$= 1.73 \times 10^{10}$

53. (*b*) $dA = 2\pi r\, dr$

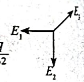
$dq = \sigma \times 2\pi r\, dr$

$$\Rightarrow E = \int_0^R \frac{kdqZ}{(Z^2+r^2)^{3/2}}$$

$$\Rightarrow E = \frac{\sigma}{2\varepsilon_0}\left[1 - \frac{Z}{\sqrt{R^2+Z^2}}\right]$$



54. (*a*) $E_1 = \dfrac{kq}{\ell^2} = E_2$



$$E_3 = \frac{kq}{(\sqrt{2}\ell)^2} = \frac{kq}{2\ell^2}$$

$$E = \frac{kq}{2\ell^2}(2\sqrt{2} - 1)$$

55. (*a*) The linear charge density of wire,

$$\lambda = \frac{-Q}{R \times 2\pi/3}$$

The electric field at the centre of curvature,

$$\vec{E} = \frac{2k\lambda}{R}\sin\left(\frac{\theta}{2}\right)(-\hat{i})$$

$$\Rightarrow \vec{E} = \frac{2k}{R} \times \frac{-3Q}{2\pi R}\sin 60°(-\hat{i})$$

$$\Rightarrow \vec{E} = \frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(-\hat{i})$$

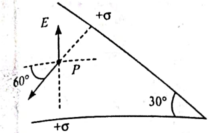
56. (*a*) Electric field due to each sheet

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}$$

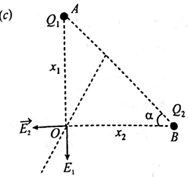
So, net electric field between plates

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\cos 60°(-\hat{x}) + \left[\frac{\sigma}{2\varepsilon_0} - \frac{\sigma}{2\varepsilon_0}\sin 60°\right](\hat{y})$$

$$\vec{E} = \frac{\sigma}{2\varepsilon_0}\left[\left(1 - \frac{\sqrt{3}}{2}\right)\hat{y} - \frac{1}{2}\hat{x}\right]$$



57. (*c*)

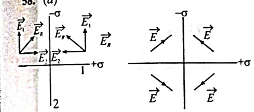


Net field along AB at O must be zero

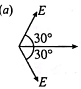
$E_2 \cos\alpha = E_1 \sin\alpha$

$$\frac{kQ_2}{x_2^2}\frac{x_2}{AB} = \frac{kQ_1}{x_1^2}\frac{x_1}{AB} \Rightarrow \frac{Q_1}{Q_2} = \frac{x_1}{x_2}$$

58. (*d*)



59. (*a*)



$E = \dfrac{4kq}{d^2}$

$$E_{net} = \frac{4kq}{d^2} \times 2\cos 30° = \frac{q\sqrt{3}}{\pi\varepsilon_0 d^2},$$

60. (*a*) For a solid sphere

$$E = \frac{\rho r}{3\varepsilon_0} \text{ for } (r < R);\ E_{1A} = \frac{-\rho R}{2(3\varepsilon_0)};|E_A| = \frac{\rho R}{6\varepsilon_0}$$

Electric field at point $B = E_B = E_{1A} + E_{2A}$

E_{1A} = Electric field due to solid sphere of radius R at point $B = \dfrac{\rho R}{3\varepsilon_0}$

E_{2A} = Electric field due to solid sphere of radius $R/2$ (which having charge density–ρ)

$$E_{2A} = -\frac{KQ' \times 4}{9R^2} = -\frac{\rho R}{54\varepsilon_0}$$

$$\left(Q' = \frac{\rho 4}{3}\pi\left(\frac{r}{2}\right)^3\right)$$

$$E_B = E_{1A} + E_{2A} = \frac{\rho R}{3\varepsilon_0} - \frac{\rho R}{54\varepsilon_0} = \frac{17\rho R}{54\varepsilon_0}$$

$$\left|\frac{E_A}{E_B}\right| = \frac{9}{17} = \frac{2 \times 9}{2 \times 17} = \frac{18}{34}$$

61. (*a*) $\vec{E} = \dfrac{kq_1}{r_1^3}\vec{r}_1 + \dfrac{kq_2}{r_2^3}\vec{r}_2$

$$= k \times 10^{-6}\left[\frac{\sqrt{10}}{10\sqrt{10}}(-\hat{i} + 3\hat{j}) + \frac{(-25)}{125}(-4\hat{i} + 3\hat{j})\right]$$

$$= (9\times10^3)\left[\frac{1}{10}(-\hat{i}+3\hat{j}) - \frac{1}{5}(-4\hat{i}+3\hat{j})\right]$$

$$= (9\times10^3)\left[\left(-\frac{1}{10}+\frac{4}{5}\right)\hat{i} + \left(\frac{3}{10}-\frac{3}{5}\right)\hat{j}\right]$$

$$= 9000\left(\frac{7}{10}\hat{i} - \frac{3}{10}\hat{j}\right)$$

$$= (63\hat{i} - 27\hat{j})(100)$$

62. (*b*) Electric field

$$E = \frac{kQh}{(h^2+R^2)^{3/2}}$$

For maxima $\dfrac{dE}{dh} = 0$

After solving we get, $\left(h = \pm\dfrac{R}{\sqrt{2}}\right)$

63. (*d*) Electric field at p

$= 2E_1\cos\theta_1 - 2E_2\cos\theta_2$

$$= \frac{2kq}{(d^2+D^2)} \times \frac{D}{(d^2+D^2)^{1/2}}$$

$$- \frac{2kq}{\left[(2d)^2+D^2\right]} \times \frac{D}{\left[(2d)^2+D^2\right]^{1/2}}$$

$$= 2kqD\left[(d^2+D^2)^{-3/2} - (4d^2+D^2)^{-3/2}\right]$$

$$= \frac{2kqD}{D^3}\left[\left(1+\frac{d^2}{D^2}\right)^{-3/2} - \left(1+\frac{4d^2}{D^2}\right)^{-3/2}\right]$$

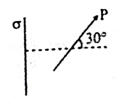
Applying binomial approximation ∵ $d << D$

$$= \frac{2kqD}{D^3}\left[1 - \frac{3}{2}\frac{d^2}{D^2} - \left(1 - \frac{3\times4d^2}{2D^2}\right)\right]$$

$$= \frac{2kqD}{D^3}\left[\frac{12}{2}\frac{d^2}{D^2} - \frac{3}{2}\frac{d^2}{D^2}\right]$$

$$= \frac{9kqd^2}{D^4}$$

64. (*b*)



The electric field E from an infinite sheet with charge density σ is

$$E = \frac{\sigma}{2\varepsilon_0}$$

The torque τ on the dipole is

$T = PE\sin\theta$

$$= \left[(2\times10^{-6})\left(\frac{2}{10}\right)\right]\left[\frac{100\times10^{-6}}{2\times8.85\times10^{-12}}\right]\left(\frac{1}{2}\right)$$

$$= \frac{10}{8.85} = 1.12 \text{Nm}$$

65. (*d*) Since polar dielectrics have dipoles oriented randomly without an external field, the net dipole moment is zero. If no field is present, polar dielectrics remain polar but their dipoles are haphazardly aligned.

66. (*a*) The electric field

$$E = \frac{V}{d} = \frac{5}{5\times10^{-4}} = 10^4 \text{ V/m}$$

is established between the plates. The dipole moment $p = q \cdot a = 2 \times 10^{-6} \times 5 \times 10^{-7} = 1 \times 10^{-12}$ C · m

Torque $\tau = pE\sin\theta$, with $\theta = 30°$ and $\sin 30° = 0.5 \Rightarrow \tau = 1 \times 10^{-12} \times 10^4 \times 0.5$ $= 5 \times 10^{-9}$ N-m.

67. (*a*) The electric potential V at a distance r from a dipole is given by:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{P}\cdot\hat{r}}{r^2}$$

Along the equatorial line (y-axis), the potential is zero.

Thus, at point B on the y-axis, $V_B = 0$

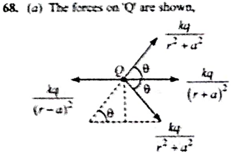
The magnitude of the electric field at the equatorial line (y-axis):

$$E_{equatorial} = \frac{p}{4\pi\varepsilon_0(r)^3}$$

At point B ($r = 2r$), the equatorial field is:

$$\text{So, } E_B = \frac{p}{4\pi\varepsilon_0(2r)^3} = \frac{E_A}{16}$$

68. (*a*) The forces on 'Q' are shown,



for $(\vec{F}_{net})Q = 0$

$$\frac{kq}{(r-a)^2} = \frac{kq}{(r+a)^2} + \frac{2kq}{(r^2+a^2)}\cos\theta$$

$$\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} = \frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

$$\frac{4ra}{(r^2-a^2)^2}=\frac{2a}{(r^2+a^2)^{\frac{3}{2}}}$$

The above equation is transcendental it could not be solved directly.

On solving this $\left|\frac{a}{r}\right|>1\Rightarrow a>r$

But point charge Q lies between charges of dipole 1 hence electric field cannot be zero.

Therefore, it should be bonus. But by solving from mathematical software we are getting a/r ≈ 3.

69. (d) $W=PE(\cos\theta_1-\cos\theta_2)$

$=2PE\ (\theta_1=0^\circ\ \theta_2=180^\circ)$

$P=4\times10^{-6}\times\frac{18}{600}$

$P=72\times10^{-8}$ cm

$W=72\times10^{-8}\times2\times10^{4}$

$\boxed{W=14.4mJ}$

70. [12] Potential energy is given by

$U=-pE\cos\theta$

$W=\Delta U=-pE(\cos\theta_f-\cos\theta_i)$

$[p=6\times10^{-6}$ cm; $E=10^6$ v/m]

$W=2pE=12$ J

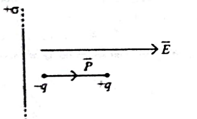
71. (c) $I\omega^2\theta=q\ell E_0\theta$ (for small 'θ')

$2m\left(\frac{\ell}{2}\right)^2\omega^2=q\ell E_0$

$\omega^2=\frac{2qE_0}{m\ell}$

$T=2\pi\sqrt{\frac{m\ell}{2qE_0}}$

72. (c)



Torque: Since the electric field is uniform and the dipole is aligned with the field, there is no torque on the dipole because the angle between the dipole moment and the electric field is zero.

Net Force: The net force on the dipole will not be zero because the electric field acts equally on both charges of the dipole, but they experience equal and opposite forces in the direction of the field, leading to a net force on the dipole directed towards the sheet.

Potential Energy: The potential energy of the dipole is minimized when the dipole is aligned with the electric field, as described by the equation $U=-\mathbf{p}.\mathbf{E}$. the correct answer is (c).

73. [2.00]

(1, 0, 4) A −4μC .... (1, 0, 4) B 4μC

As, $\vec p=q\vec\ell$ and $\vec\tau=\vec p\times\vec E$

$\vec E=0.2\frac{V}{cm}=20\frac{V}{m}$

$\vec p=4\times(\hat i-\hat j+\hat k)$

$=(4\hat i-4\hat j+4\hat k)\mu C-m$

$\vec\tau=(4\hat i-4\hat j+4\hat k)\times(20\hat i)\times10^{-6}\,Nm$

$=(8\hat k+8\hat j)\times10^{-5}=8\sqrt2\times10^{-5}$

α = 2

74. [16] Electric field at point P

$E_P=\frac{2KP}{r^3}=E$

Electric field at point R,

$E_R=\frac{KP}{(2r)^3}=\frac{E}{16}$

$x=16$

75. (c) Electric field due to a dipole at point on its equatorial plane

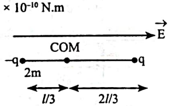
$=\frac{1}{4\pi\epsilon_0}\frac{2p}{r^3}\Rightarrow E\propto\frac{1}{r^3}$

76. (b) $|\vec\tau|=PE\sin\theta=(q.d)E\sin\theta$

$=0.01\times0.4\times10^{-3}\times10\times10^{-5}\times\sin30^\circ$

$=2\times10^{-10}$ N.m

77. (*)



If released, it will oscillate about centre of mass(COM).

$\tau=-pE.\sin\theta=-pE.\theta$ {∵ θ is small}

$\Rightarrow\left[2m\frac{l^2}{9}+m\frac{4l^2}{9}\right]\alpha=-qlE.\theta$

$\Rightarrow\frac{2ml^2}{3}\alpha=-qlE\cdot\theta\Rightarrow\alpha=-\frac{3qE}{2ml}\theta$

∴ Angular frequency

$\omega=\sqrt{\frac{3qE}{2ml}}$ (Bonus)

78. (a) Force on charges will be different in nonuniform field.

79. [6] $|\tau|_{max}=PE$

So,

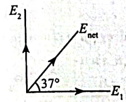
$\frac{\tau_1}{\tau_2}=\frac{P_1E_1}{P_2E_2}=\frac{(1.2\times10^{-30})\times(5\times10^4)}{(2.4\times10^{-30})\times(15\times10^4)}=\frac{1}{6}$

$\therefore\frac{1}{6}=\frac{1}{x}\Rightarrow x=6$

80. (d) Electric field at an axial line of a dipole, $E_1=k\frac{2p_1}{r^3}$

Electric field at on equatorial line of a dipole, $E_2=k\frac{p_2}{r^3}$

$\tan37^\circ=\frac{E_2}{E_1}=\frac{p_2}{2p_1}=\frac{3}{4}$



$\frac{p_1}{p_2}=\frac{2}{3}$

81. (d) $|\vec F|=q(|\vec E_1|-|\vec E_2|)=2k\lambda q\left[\frac{1}{r_1}-\frac{1}{r_2}\right]$

$|\vec F|=2\times9\times10^9\times3\times10^{-6}\left[\frac{1000}{10}-\frac{1000}{12}\right]q$

$4=9\times10^5\,q$

$q=\frac{4}{9\times10^5}=4.44\mu C$

82. (a) Since, $\vec p.\vec r=0$

$\vec E$ must be antiparallel to $\vec p$

So, $\vec E=-\lambda(-\vec p)$

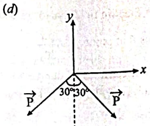
where λ is an arbitary positive constant

Now $\vec A=a\hat i+b\hat j+c\hat k$

$\vec A\parallel\vec E$

$\frac{a}{\lambda}=\frac{b}{3\lambda}=\frac{c}{-2\lambda}=k$

83. (d)

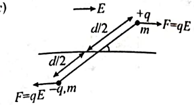


$|\vec P|=ql$

$\vec P_{net}=2P\cos30^\circ(-\hat j)=P\sqrt3(-\hat j)$

$=\sqrt3\,ql(-\hat j)=-\sqrt3\,q\ell\,\hat j$

84. (c)



moment of inertia $(I)=m\left(\frac{d}{2}\right)^2\times2=\frac{md^2}{2}$

Now by $\tau=I\alpha$

$(qE)(d\sin\theta)=\frac{md^2}{2}\cdot\alpha$

$\alpha=\left(\frac{2qE}{md}\right)\sin\theta$

for small θ, sin θ ≈ θ

$\Rightarrow\alpha=\left(\frac{2qE}{md}\right)\theta\Rightarrow w=\sqrt{\frac{2qE}{md}}$

85. (a) Given Data:

Length of the plates:

L = 10 cm = 0.1 m

Initial horizontal velocity: $v_x=10^6$ m/s

Electric field: E = 9.1 V/ cm = 910 V/m

Charge of an electron: $e=1.6\times10^{-19}$C

Mass of an electron: $m=9.1\times10^{-31}$ kg

The force on the electron due to the electric field is given by:

$F=eE=(1.6\times10^{-19})\times(910)=1.456\times10^{-16}$ N

$a=\frac{F}{m}=\frac{1.456\times10^{-16}}{9.1\times10^{-31}}$

$a=1.6\times10^{14}$ m/s²

The time t taken by the electron to travel through the plates is given by:

$t=\frac{L}{v_x}=\frac{0.1}{10^6}=10^{-7}$s

Since the electron is initially moving horizontally, its initial vertical velocity is 0. The vertical velocity is given by:

v_y = at

$v_y=(1.6\times10^{14})\times(10^{-7})$

$v_y=1.6\times10^7$ m/s

86. (b) From work energy theorem

Work by Electric force = $W(F_e)$

$=\frac{1}{2}mv^2-0$

$qE\frac{l}{2}=\frac{1}{2}mv^2$

$=\sqrt{\frac{qEl}{m}}$

87. (b) $\frac{2k\lambda q}{r}=m\omega^2r$

$\omega^2=\frac{2k\lambda q}{mr^2}$

$\left(\frac{2\pi}{T}\right)^2=\frac{2k\lambda q}{mr^2}$

$T=2\pi r\sqrt{\frac{m}{2k\lambda q}}$

88. [8] $u=1$ m/s; $a=-\frac{\sigma e}{2\varepsilon_0m}$

$t=1$ s

$S=-1$ m

Using $S=ut+\frac{1}{2}at^2$

$-1=1\times1-\frac{1}{2}\times\frac{\sigma e}{2\varepsilon_0m}\times(1)^2$

$\therefore\sigma=8\frac{\varepsilon_0m}{e}\times(1)^2$

∴ α = 8

89. [45] ∵ $0.5e=\frac{1}{2}mv_x^2$

$\Rightarrow v_x=\sqrt{\frac{e}{m}}$

Along x axis, $L=v_xt=\sqrt{\frac{e}{m}}t$ ...(i)

Along y axis $v_y=\frac{eE}{m}t$ ...(ii)

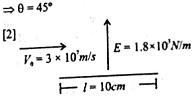
$\frac{v_y}{L}=E\sqrt{\frac{e}{m}}=Ev_x$

∵ $\tan\theta=\frac{v_y}{v_x}$

$\Rightarrow\tan\theta=E\times L=10\times0.1=1$

$\Rightarrow\theta=45^\circ$

90. [2]



$\because a=\frac{qE}{m}=(2\times10^{11})(1.8\times10^3)$

$=3.6\times10^{14}\ m/s^2$

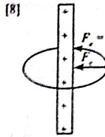
Time to cross plates $=\frac{d}{v}$

$\Rightarrow t=\frac{0.10}{3\times10^7}$

$y=\frac{1}{2}at^2=\frac{1}{2}(3.6\times10^{14})\left(\frac{0.01}{9\times10^{14}}\right)$

$=0.2\times0.01\Rightarrow\approx2$ mm

91. [8]



Centripetal force is provided by electric field.

$F_e=F_c$

$eE=\frac{mv^2}{r}\Rightarrow e\cdot\frac{2k\lambda}{r}=\frac{mv^2}{r}$

$V=\sqrt{\frac{e\cdot2k\lambda}{m}}$

$=\sqrt{\frac{1.6\times10^{-19}\times2\times9\times10^9\times2\times10^{-8}}{9\times10^{-31}}}$

$=\sqrt{64\times10^{12}}$

92. (b) qE = mg

$\Rightarrow q=\frac{mg}{E}=2\times10^{-9}$C.

93. (b) We have, $E=\frac{8m}{e}$ v/m

time to come out, $t=\frac{1}{2}$ sec

$V_y=at=\left(\frac{eE}{m}\right)t=\left(\frac{e}{m}\times\frac{8m}{e}\right)\times\frac{1}{2}=4$ m/s

So, $\tan\theta=\frac{V_y}{V_x}=\frac{4}{2}\Rightarrow\theta=\tan^{-1}2$

94. (a) Electric field at the surface of cylinder $=\frac{\delta R}{2\varepsilon_0}$

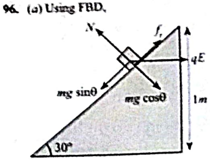
$\therefore qE=\frac{mv^2}{R}$

$\Rightarrow q\times\frac{\rho R}{2\varepsilon_0}=\frac{mv^2}{R}\Rightarrow mv^2=\frac{\rho qR^2}{2\varepsilon_0}$

$\therefore K.E=\frac{\rho qR^2}{4\varepsilon_0}$

95. (d) $S=\frac{v^2}{2a}=\frac{v^2}{2\left(\frac{qE}{m}\right)}$

96. (a) Using FBD,



$F = mg\sin\theta - qE\cos\theta - \mu N$

Here $N = mg\cos\theta + qE\sin\theta$

$$\therefore F = \frac{9.8}{2} - \frac{5\times10^{-3}\times200\times\sqrt{3}}{2}$$

$$-0.2\left(\frac{5\times10^{-3}\times200}{2} + \frac{1\times9.8\times\sqrt{3}}{2}\right)$$

$F = 2.25\ N$

$\Rightarrow a = 2.25\ \text{m/s}^2$

$S = ut + \frac{1}{2}at^2$

$\Rightarrow t = \sqrt{\frac{2l}{a}} = 1.31\,\text{sec}$

97. [1] Electric force acting on the body, $F = qE$

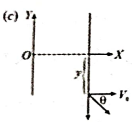
Accleration of body,

$a = \frac{F}{m} = \frac{q}{m}E = \frac{8\times10^{-6}}{10^{-3}}\times100 = 0.80\text{m/s}^2$

∴ Time period of Motion,

$T = 2\sqrt{\frac{l}{a}} = 2\sqrt{\frac{2\times0.1}{0.8}} = 1\text{s}$

98. (c)



$\frac{qE}{m}\ \frac{d}{V_0}$

$y = \frac{1}{2}\left(\frac{qE}{m}\right)\left(\frac{d}{V_0}\right)^2$

$y = -\frac{qEd}{mV_0^2}x + C$

At $x = d$,

$y = -\frac{qEd^2}{2mV_0^2} \Rightarrow C = \frac{qEd^2}{2mV_0^2}$

$y = -\frac{qEd}{mV_0^2}x + \frac{qEd^2}{2mV_0^2}$

$y = \frac{qEd}{mV_0^2}\left(\frac{d}{2} - x\right)$

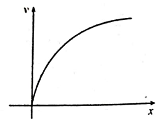
99. (a) $E = E_0(1 - ax^2)$

$\frac{vdv}{dx} = \frac{qE_0}{m}(1 - ax^2)$

$\Rightarrow \frac{v^2}{2} = \frac{qE_0}{m}\left[x - \frac{ax^3}{3}\right] = 0$

$\Rightarrow x = \sqrt{\frac{3}{a}}$

100. (d) $v^2 = \frac{2qE}{m}x$



101. (b) $\vec{F} = qE\hat{i} + mg\hat{j}$

Since initial velocity is zero. It will move in straight line.

102. (d) The wire passes through two maximally displaced (diagonally opposite) corners of the cube. The length of the wire segment inside the cube is the **space diagonal** of the cube:

$L = \sqrt{3}a = \sqrt{3}\times\sqrt{3}\times10^{-2} = 3\times10^{-2}$ m

Enclosed Charge (q_{enc}):

$q_{enc} = \lambda L = 2\times10^{-9}\times3\times10^{-2}$

$= 6\times10^{-11}$C.

Using Gauss's Law:

$\phi = \frac{q_{enc}}{\epsilon_0}$

$$\frac{6\times10^{-11}}{\frac{1}{4\pi\times9\times10^9}} = 2.16\pi\text{Nm}^2\text{C}^{-1}$$

103. [15] $\vec{E} = (2\hat{i} + 4\hat{j} + 6\hat{k})\times10^3$ N/C

For x-z plane, $\vec{A} = A\hat{j}$,

flux $\phi = \vec{E}\cdot\vec{A} = (4\times10^3)\cdot A\cdot$

$\phi = 6\times10^6\text{Nm}^2/\text{C}$,

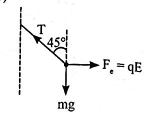
$A = 1.5\times10^{-3}\ \text{m}^2 = 15\ \text{cm}^2$

104. (b) Two infinite parallel plates with surface charge densities $+\sigma$ and $-\sigma$ are separated by distance d. A point charge $+q$ is at the midpoint. The electric field due to the $+\sigma$ plate is $\frac{\sigma}{2\epsilon_0}$ toward the $-\sigma$ plate, and due to the $-\sigma$ plate is $\frac{\sigma}{2\epsilon_0}$ toward the $+\sigma$ plate. The net field is $\frac{\sigma}{\epsilon_0}$ so the force on $+q$ is $F = q.\frac{\sigma}{\epsilon_0}$.

Plate 1	Plate 2
$\frac{-\sigma}{2} \mid \frac{3\sigma}{2}$	$\frac{-3\sigma}{2} \mid \frac{-\sigma}{2}$

$\therefore F_{net} = \frac{3\sigma}{2\epsilon_0}q$

105. (b)



At equilibrium, F = mg tan 45° = mg.

Electric field $E = \frac{\sigma}{2\epsilon_0}$, so qE = mg.

we get

$\sigma = \frac{2\epsilon_0 mg}{q}$

$= \frac{2\times8.85\times10^{-12}\times0.0001\times10}{10\times10^{-6}}$

= 1.77×10^{-9} c/m²

106. (d) since the line is at the center of the edge BC, the part of this line enclosed within the cube is only one-fourth of the total length. Thus, the part of the line charge enclosed in the cube is:

$(a/2).(1/4) = a/8$

Charge enclosed $= \frac{\lambda\frac{a}{2}}{4} = \frac{\lambda a}{8}$

$\therefore \phi = \frac{q_{in}}{\varepsilon_0} = \frac{\lambda a}{8\varepsilon_0}$

107. [48] The total electric flux through the entire square is given by:

Total flux $= \frac{q}{6\epsilon_0}$

Flux Through the-Shacted Region: Since the flux is equally divided, the flux through the shaded part is $\frac{5}{8}$ of the total flux.

Final Flux Calculation:

Flux through shaded region

$= \frac{5}{8}\times\frac{q}{6\epsilon_0} = \frac{5q}{48\epsilon_0}$

Substituting the value of $q = 1C$

$\frac{5}{48\epsilon_0}$

108. (d) Formulae based

109. (c) According to Gauss's Law,

$\Phi = \frac{q}{\epsilon_0}$

or, $q = \Phi\times\epsilon_0$

$q = (-2\times10^4)\times(8.85\times10^{-12})$

$q = -17.7\times10^{-8}\ C = -17.7\times10^{-8}\ C$

Option (c) is correct.

110. (a) $\phi_{\text{bigger cube}} = \frac{5Q}{\epsilon_0}$

$\phi_{\text{smaller cube}} = \frac{2Q}{\epsilon_0}$

$\Rightarrow \frac{\phi_{\text{smaller cube}}}{\phi_{\text{bigger cube}}} = \frac{2}{5}$

111. (c) $\vec{E} = 6\hat{i} + 5\hat{j} + 3\hat{k}$

$\vec{A} = 30\hat{i}$

Electric flux through a surface area lying in YZ-plane

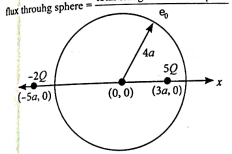
$\phi = \vec{E}.\vec{A}$

$\phi = (6\hat{i} + 5\hat{j} + 2\hat{k}).(30\hat{i})$

$\phi = 6\times30 = 180$

112. (b)

flux throuhg sphere $= \frac{\text{total charge enclosed in sphere}}{\epsilon_0}$



5Q charge is inside the spherical region flux through sphere $= \frac{5Q}{\epsilon_0}$

113. (2) Electric field E at a distance r due to infinite long wire is $F = \frac{2k\lambda e}{r}$

$F = e\left(\frac{2k\lambda}{r}\right)$ [As, F = eE]

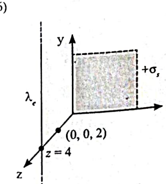
$F = \frac{2k\lambda e}{r}$.

Now, $F = \frac{mv^2}{r} = \frac{2k\lambda e}{r}$

$\Rightarrow v = \sqrt{\frac{2k\lambda e}{m}}$

$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{2k\lambda e}{m}\right) = k\lambda e$

114. (16)



$\frac{E_s}{E_t} = \frac{\sigma}{2\epsilon_0}\times\frac{2\pi\epsilon_0 r}{\lambda} = \frac{\pi\times\sigma r}{\lambda}$

$= \frac{\pi\times2\lambda\times2}{\lambda} = \frac{4\pi}{1}$

$\therefore n = 16$

115. (b) From Gauss's Law

Cover the charge completely by another identical cube. flux through each = ϕ

$2\phi = \frac{q}{\epsilon_0}$

$\phi = \frac{q}{2\epsilon_0}$

116. (2) From Gauss's law

$E'A = \frac{\sigma\times A}{\epsilon_0} \Rightarrow E = \frac{r}{\epsilon_0}$

117. [12] Electric flux for the surface,

$\phi = \vec{E}\cdot\vec{A}$

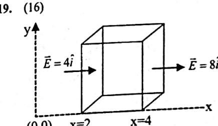
$= \left(\frac{2\hat{i} + 6\hat{j} + 8\hat{k}}{\sqrt{6}}\right)\cdot4\left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}\right)$

$= \frac{4}{6}\times(4 + 6 + 8) = 12\text{Vm}$

118. (b) From gauss law electricflux through a surface,

$\phi = \frac{q_{in}}{\epsilon_0} = \frac{q + (-2q) + 5q}{\epsilon_0} = \frac{4q}{\epsilon_0}$

119. (16)



$\bar{E} = 2x\hat{i}$

$\phi = \bar{E}\cdot\bar{A}$

$\phi_{in} = -4\times4 = -16\text{Nm}^2/\text{c}$

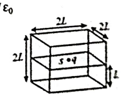
$\phi_{out} = 8\times4 = 32\ \text{Nm}^2/\text{c}$

The net electric flux through the cube,

$\phi_{net} = \phi_{in} + \phi_{out}$

$= -16 + 32 = 16\ \text{Nm}^2/\text{c}$

120. (d) Total electric flux through the cube $\phi = q/\epsilon_0$



Flux passing through the face opposite to surface S $= \frac{q}{6\epsilon_0}$

121. [640] Electric Flux $= \vec{E}\cdot\vec{A}$

$= 4000(0\cdot2)^2\times(0\cdot2)^2$

$= 4000\times16\times10^{-4}$ Vm

= 640 V cm

122. (*) ∴ Flux through curve surface of hemisphere $\phi = \frac{q}{2\epsilon_0}$

Note: Flux through flat surface should be zero, but option is not given. Hence it is bonus.

123. (c) Total flux through the cube $= (q/\epsilon_0)\times1/8 = q/8\epsilon_0$

Total flux through one "outer" face of the cube $= (q/8\epsilon_0)\times1/3 = q/24\epsilon_0$

[Because there is flux only through 3 faces]

Hence, total flux through shaded area

$\phi_T = (q/25\epsilon_0 + q/24\epsilon_0)\times\frac{1}{2}$ [half of each face is shaded]

$\phi_T = q/24\epsilon_0$

124. (a) We know that a dipole is a pair of equal and opposite charges separated by a small distance.

∵ The net charge on the electric dipole,

$q_{net} = +q - q = 0$

Therefore, according to Gauss's law,

Electric flux, $\oint\vec{E}\cdot\vec{ds} = \frac{q_{net}}{\epsilon_0} = \frac{0}{\epsilon_0} = 0$

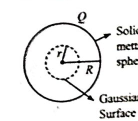
Where, $\vec{E}$ is the electric field, $\vec{ds}$ is the surface area a small section, and ϵ_0 is permittivity in free space.

And for a small section $\vec{ds}$ only, $\vec{E}\cdot\vec{ds} \neq 0$

∵ The flux of the electric field throughout the hollow sphere is zero and the electric field is non-zero.

Hence, statement-I is true :

Electric field in region ($r < R$)



$\vec{E} = \frac{1}{4\pi\epsilon_0}\times\frac{Qr}{R^3}$

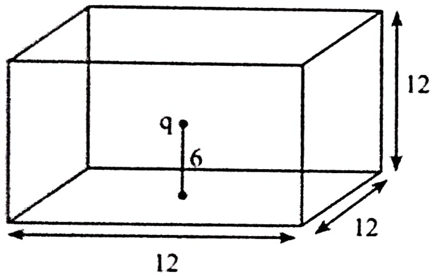
$\therefore$ The electric field due to the charged solid sphere at a distance r from the centre is non-zero.

As change encloses within the gaussian surface is equal to zero such as.

$\phi = \oint \vec{E} \cdot \vec{ds} = 0$

Hence, Statement-II is false:

125. [226]



$$\phi = \frac{q}{6\varepsilon_0} = \frac{12\mu C}{6\varepsilon_0} = 226 \times 10^3\ Nm^2/c$$

126. [4] Using gauss law

$$\vec{\nabla}.\vec{E} = \frac{\rho}{\varepsilon_0}$$

$$\Rightarrow \vec{\nabla}.(\varepsilon_0 \vec{E}) = \rho$$

$$\Rightarrow \vec{\nabla}.\vec{D} = \rho$$

$$\Rightarrow \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right).(e^{-x}\sin y\hat{i} - e^{-x}\cos y\hat{j} + 2z\hat{k}) = \rho$$

$$\Rightarrow \rho = -e^{-x}\sin y + e^{-x}\sin y + 2$$

At origin $x = 0, y = 0$

$\therefore e = -0 + 0 + 2 = 2$

Hence charge $= eV$

$= 2\ (2 \times 10^{-9}) = 4 \times 10^{-9} = 4$ nC

127. [1] $$\frac{\phi_1}{\phi_2} = \frac{\frac{3}{5}E_0 \times 0.2}{\frac{4}{5}E_0 \times 0.3} = \frac{1}{2}$$

$\Rightarrow a = 1$

128. [640] $\phi = \left(\frac{2}{5}E_0 \times 0.4\right) = 640\ Nm^2/C$

129. [−48] Flux via $ABCD$

$$\phi_1 = \int \vec{E}.d\vec{A} = 0$$

Flux via BCEF

$$\phi_2 = \vec{E}.\vec{A} = (4x\hat{i} - (y^2+1)\hat{j}).4\hat{i} = 16x\ (\text{At } x = 3)$$

$$\phi_2 = 48\frac{N.m^2}{C};$$

$$\phi_1 - \phi_2 = -48\frac{N.m^2}{C}$$